Solution:

$p = \frac{4}{x} \Rightarrow x = \frac{4}{p}$

$\therefore \frac{dx}{dp} = -\frac{4}{p^2}$

At $p = 4$, $\frac{dx}{dp} = -\frac{1}{4} = -0.25$

$\therefore$ Rate of change of demand with respect to the price at $p =$ ₹ 4 is -0.25.

Interpretation:

When the price increases by 1% from the level of $p =$ ₹ 4, the demand decreases (falls) by 0.25%.

Example 6.16

The demand and the cost function of a firm are $p = 497 - 0.2x$ and $C = 25x + 10000$ respectively. Find the output level and price at which the profit is maximum.

Solution:

We know that profit[P(x)] is maximum when marginal revenue [MR] = marginal cost [MC].

Revenue: $R = px$

$= (497 - 0.2x)x = 497x - 0.2x^2$

$MR = \frac{dR}{dx} = 497 - 0.4x$

Cost: $C = 25x + 10000$

$\therefore MC = 25$

$MR = MC \Rightarrow 497 - 0.4x = 25$

$\Rightarrow 472 - 0.4x = 0$

$\Rightarrow x = 1180$ units.

Now, $p = 497 - 0.2x$

at $x = 1180$, $p = 497 - 0.2(1180) =$ ₹ 261.

Example 6.17

The cost function of a firm is $C = \frac{1}{3}x^3 - 3x^2 + 9x$. Find the level of output ($x > 0$) when average cost is minimum.

Solution:

We know that average cost [AC] is minimum when average cost [AC] = marginal cost [MC].

Cost: $C = \frac{1}{3}x^3 - 3x^2 + 9x$

$\therefore AC = \frac{1}{3}x^2 - 3x + 9$ and $MC = x^2 - 6x + 9$

Now, $AC = MC \Rightarrow \frac{1}{3}x^2 - 3x + 9 = x^2 - 6x + 9$

$\Rightarrow 2x^2 - 9x = 0 \Rightarrow x = \frac{9}{2}$ units. [$\because x > 0$]

Exercise 6.1

1. A firm produces x tonnes of output at a total cost of $C(x) = \frac{1}{10}x^3 - 4x^2 - 20x + 7$. Find the
   (i) average cost function
   (ii) average variable cost function
   (iii) average fixed cost function
   (iv) marginal cost function and
   (v) marginal average cost function.
2. The total cost of x units of output of a firm is given by $C = \frac{2}{3}x + \frac{35}{2}$. Find the
   (i) cost, when output is 4 units
   (ii) average cost, when output is 10 units
   (iii) marginal cost, when output is 3 units
3. Revenue function 'R' and cost function 'C' are $R = 14x - x^2$ and $C = x\left(x^2 - 2\right)$. Find the (i) average cost function (ii) marginal cost function, (iii) average revenue function and (iv) marginal revenue function.
4. If the demand law is given by $p = 10e^{-\frac{x}{2}}$, then find the elasticity of demand.

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